

PUJA MINODJI THAKRE

Mathematics Researcher • Knowledge Representation • Statistical Learning & Data Mining

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RESEARCH INTERESTS

Knowledge Representation and Knowledge Graphs; Machine Learning and Data Mining; Natural Language Processing; Statistical Learning; Mathematical Modelling; Intelligent Information Systems.

EDUCATION

Indian Institute of Technology Delhi (IIT Delhi)

M.Sc. Mathematics

2023–2025

CGPA: 7.65/10

Government Institute of Science, Nagpur

B.Sc.

2019–2022

91.70%

RESEARCH EXPERIENCE

Cryptocurrency Markets and Geopolitical Risk: A Study of the Israel–Gaza Conflict

2025–Present

Collaborative research; replication and extension of a high-frequency event-study framework

- Collaborated on a quantitative study examining the response of **cryptocurrency and traditional financial markets** to geopolitical shocks during the Israel–Gaza conflict.
- Contributed to the construction of an **Hourly War Risk Index (HWRI)** from time-stamped geopolitical news to distinguish escalation and de-escalation shocks.
- Integrated geopolitical events with hourly financial data, consolidating **3,395 reports into 2,835 unique events**; estimated **AR, CAR and CAAR** across 1–48 hour event windows.
- Results indicated **heterogeneous and short-lived asset-specific responses**, with no persistent market-wide relationship between war risk and abnormal returns over the full study period.

Knowledge Representation for Healthcare Systems: A Comparative Analysis of Performance of Different Representation Paradigms

2024–2026

Collaborative research; published in Springer proceedings

- Contributed to a unified framework comparing **knowledge-representation paradigms** for healthcare question answering across increasing levels of reasoning complexity.
- Worked with a benchmark of **47 cardiovascular patient records** and five clinical queries spanning Boolean, relational, multi-relational and contextual reasoning.
- Contributed to comparative evaluation using **query coverage, reasoning depth and explainability**; Knowledge Graphs achieved **complete benchmark coverage** with multi-hop and contextual reasoning capabilities.

PUBLICATION

Atul Kumar Tripathi, **Puja Minodji Thakre**, and Niladri Chatterjee.

Knowledge Representation for Healthcare Systems: A Comparative Analysis of Performance of Different Representation Paradigms.

Intelligent Computing: Proceedings of the 2026 Computing Conference, Volume 2, Lecture Notes in Networks and Information Systems, vol. 1950, pp. 38–54, Springer, 2026.

DOI: 10.1007/978-3-032-24807-7_4

ACADEMIC DISTINCTIONS

- **IIT JAM Mathematics (2023)** — All India Rank **237**.
- **University Rank Holder** — Secured the **4th-highest rank** at Rashtrasant Tukadoji Maharaj Nagpur University in B.Sc.
- **Gold Medal & Cash Prize** — Awarded for securing the **highest marks in Physics and Mathematics** in B.Sc.
- **Shri V. S. Tamma Silver Medal** — Awarded for securing the **highest marks in Physics** in B.Sc.

SCHOLARSHIPS & ACADEMIC LEADERSHIP

- **National Scholarship Portal (NSP) Scholarship** — Awarded INR 3,00,000 during M.Sc. based on academic performance in graduation.
- **AASCON Scholarship** — Awarded for securing the **highest marks and first position** in B.Sc. examinations.
- **University Rank Holder** — Secured the **4th-highest rank** in B.Sc. at RTMNU University.

- **PG Coordinator, Office of Career Services, IIT Delhi (2024–2025)** — Coordinated postgraduate recruitment activities involving **800+ companies, 1,200+ students and 25 disciplines**, including outreach and onboarding of **40+ new firms**.

TECHNICAL SKILLS

Programming & Data Analysis: Python, SQL, C, C++, LaTeX

Machine Learning & Data Mining: Machine Learning, Data Mining, Regression, Classification, Clustering

Natural Language Processing & Deep Learning: Natural Language Processing, Text Classification, Deep Learning, Neural Networks

Knowledge Representation: Knowledge Graphs, Ontologies, RDF, OWL, SPARQL

SELECTED COURSEWORK

M.Sc. Mathematics, IIT Delhi

- **Analysis:** Real Analysis; Complex Analysis; Measure and Integration; Functional Analysis
- **Algebra & Topology:** Abstract Algebra; Linear Algebra; Topology
- **Differential Equations & Numerical Methods:** Ordinary Differential Equations; Partial Differential Equations; Numerical Analysis
- **Probability, Statistics & Optimisation:** Probability and Statistics; Mathematical Programming; Multiple Decision Procedures in Ranking and Selection
- **Computational & Data Methods:** Computer Programming; Mathematics Behind Machine Learning; Data Mining

SELECTED ACADEMIC PROJECTS

Knowledge Representation Schemes: Semantic Net to Knowledge Graph

2024–2025

M.Sc. Research Project, Department of Mathematics, IIT Delhi

Supervisor: Prof. Niladri Chatterjee, Former Professor, IIT Delhi

- Studied the evolution of **knowledge representation** from formal logic and semantic networks to graph-based representations, with emphasis on structured and relational knowledge.
- Explored **Knowledge Graph development** through graph data models, RDF/RDFS, triples, classes, properties and semantic relationships.
- Implemented and examined knowledge representation using **RDF and SPARQL**, focusing on structured knowledge modelling and query-based information retrieval.

PROFESSIONAL EXPERIENCE

Resonance, Hyderabad

July 2025–Present

Mathematics Faculty — JEE & Olympiad Programs

- Teach Mathematics for **JEE Main, JEE Advanced and Olympiad Foundation** programs, with emphasis on advanced problem solving and mathematical reasoning.
- Develop and review **mathematical problems, assessments and instructional material** for classroom and digital learning programs.

ACADEMIC SERVICE & ENGAGEMENT

NSS Volunteer, 2019–2022 — Community-service initiatives in women’s empowerment, environmental awareness, and educational outreach.

Cultural Achievements — Winner, college dance competition; Second Place, inter-college fashion show.

REFERENCES

Dr. Niladri Chatterjee

Professor & Research Supervisor

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Dr. Neeraj Joshi

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Prof. Sudip Bapat

Assistant Professor & Research Supervisor

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